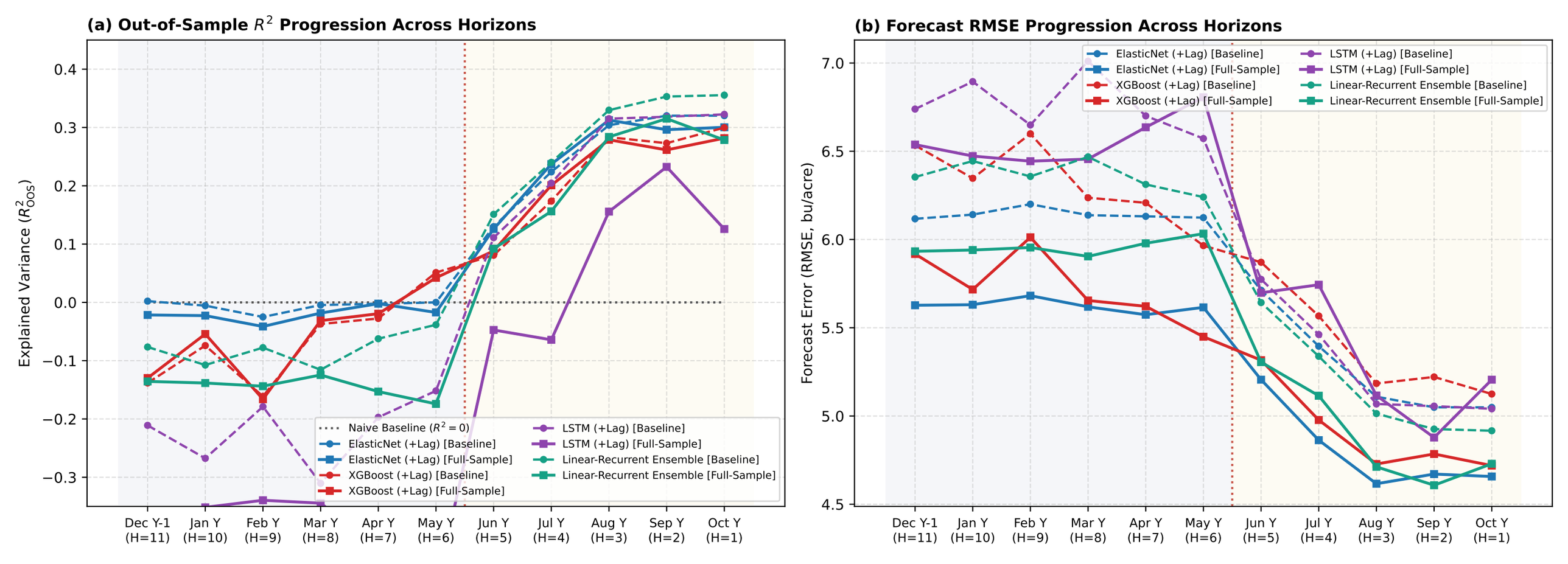
(a) Out-of-Sample R^2 Progression Across Horizons
Explained Variance (R^2_{OOS})
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Naive Baseline ($R^2 = 0$)
ElasticNet (+Lag) [Baseline]
ElasticNet (+Lag) [Full-Sample]
XGBoost (+Lag) [Baseline]
XGBoost (+Lag) [Full-Sample]
LSTM (+Lag) [Baseline]
LSTM (+Lag) [Full-Sample]
Linear-Recurrent Ensemble [Baseline]
Linear-Recurrent Ensemble [Full-Sample]
Dec Y-1 (H=11)
Jan Y (H=10)
Feb Y (H=9)
Mar Y (H=8)
Apr Y (H=7)
May Y (H=6)
Jun Y (H=5)
Jul Y (H=4)
Aug Y (H=3)
Sep Y (H=2)
Oct Y (H=1)
(b) Forecast RMSE Progression Across Horizons
Forecast Error (RMSE, bu/acre)
7.0
6.5
6.0
5.5
5.0
4.5
ElasticNet (+Lag) [Baseline]
ElasticNet (+Lag) [Full-Sample]
XGBoost (+Lag) [Baseline]
XGBoost (+Lag) [Full-Sample]
LSTM (+Lag) [Baseline]
LSTM (+Lag) [Full-Sample]
Linear-Recurrent Ensemble [Baseline]
Linear-Recurrent Ensemble [Full-Sample]
Dec Y-1 (H=11)
Jan Y (H=10)
Feb Y (H=9)
Mar Y (H=8)
Apr Y (H=7)
May Y (H=6)
Jun Y (H=5)
Jul Y (H=4)
Aug Y (H=3)
Sep Y (H=2)
Oct Y (H=1)